Hauke Sandhaus

Work Portfolio

Unpublished work

Volkswagen Group Future Center GmbH

Vision: Mobility for ALL, at the push of a button

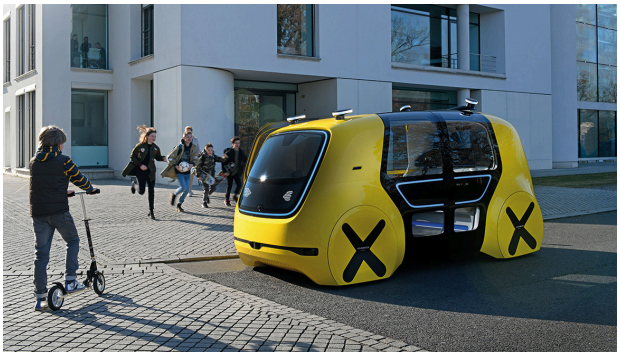
Department: Digital User Experience Design

The Future Center is an independent organization in the Volkswagen Group. It develops concepts for the many brands of the group. The UX department in the Future Center has been working on concepts for autonomous driving since its installation.

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Self-driving cars

#inclusive_design #urban_transformation



The main goal was to develop the user experience of self-driving cars to enable our vision. Fully autonomous pods represent a sharp break from current vehicles; the relationship to users and the environment requires a rethinking of even the basics. Self-driving cars should provide better, more sustainable, and more accessible options for all.

I, as a generalist, was a member of the strategy, research, and software circle but also created concepts. The concepts were always realized in tangible prototypes in order to verify them with users and turn them into requirements.

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Flying cars

#0.1% #drones #augmented_reality #semi-manual-flight



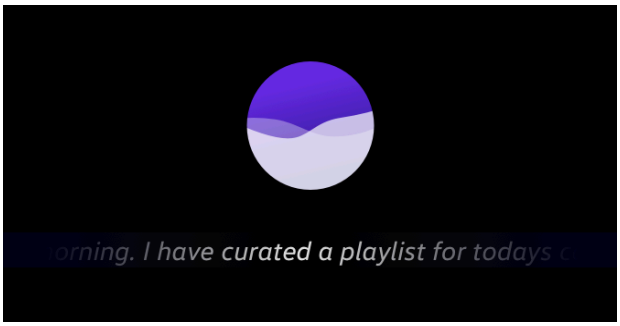
Created a lightweight HMI concept for an electric vertical takeoff and landing vehicle from a German luxurious car manufacturer.

The concept was based on needs collected from interviews of super high net worth individuals.

I was responsible for the user need analysis, creation of hardware for controlling the vehicle, and development of augmented reality interactions in a flight simulator.

AI Voice

#ANN #ML #voice_interaction #privacy #assistant #visual_AI



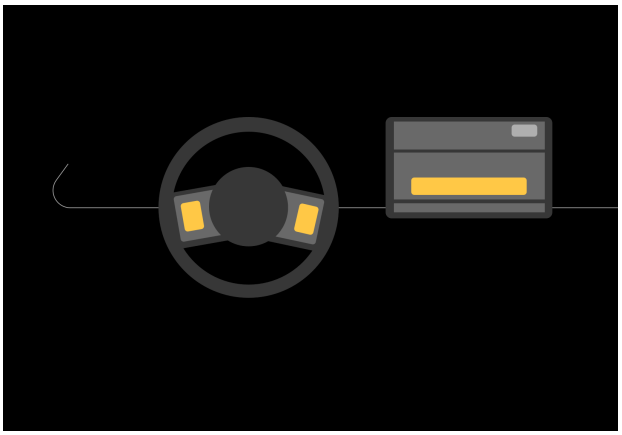
Created an emphatic voice assistant and voice interactions for fully and semi-autonomous vehicles. Worked with visually impaired & older adults.

Developed proactive use cases enabled by visual AI with respect to passengers' privacy.

I was responsible for the voice recognition with embedded hardware (Snips AI), training of models, and creation of software prototypes with machine learning models (OpenPose, YOLO & runway.ml). Furthermore, I created motorized camera shutters with respect to in-cabin privacy and tested the voice feedback of autonomous vehicles for the visually impaired.

Digital Driver UI

front-end-development #measuring_UX #ideation



Worked on concepts for innovative car interfaces. Enabling productivity and entertainment in new fully connected vehicles with increasing automation and features. These concepts were realized with high-fidelity interactive prototypes, completely explorable for users, with the intention to bring them to production.

I was responsible for the ideation, implementation of the front-end software & hardware as well as integration with a driving simulator. Furthermore I, as the research link, was organizing and conducting qualitative and quantitative research.

Personal Portfolio

This pdf presents only a high-level overview of my current work portfolio. For my public portfolio consult:

http://www.haukesandhaus.de/